



ADITYA YADAV *AI Engineer*

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📄 SUMMARY

AI engineer with 3 years of experience building production agent systems end-to-end. I design agent harnesses: routing, tool boundaries, execution state, and deciding what context an agent gets at which step, which is where most of the hard problems live once a system leaves the demo. I build the retrieval and document pipelines underneath in Python with FastAPI, and I evaluate whether agents actually complete their tasks rather than judging them by how good the output looks.

📁 WORK EXPERIENCE

Full Time AI Developer, Sogedes GmbH

12/2024 – 04/2026 | Mannheim, Germany

- **AI Platform & Backend Architecture (AiFctry):** Core contributor to AiFctry, an enterprise AI platform powering Voice Bot, Chatbot and Agent Assist products for large enterprise clients, including a major European telecommunications provider. Built proof-of-concept prototypes with stakeholders and shipped the ones that proved out into production. Developed containerised, cloud-native microservices (FastAPI, Docker), benchmarked models across OpenAI, Groq, and Gemini, and built evaluation scripts to measure quality, latency, and cost.
- **AI Agent Assist & Agent Orchestration:** Engineered a LangGraph-based Next-Step Recommender on the LangChain ecosystem, using tool calling and a state-machine framework to define the agent harness: which tools the agent could reach for at which point, how execution state carried across multi-step actions, and what happened when a step failed. Built separate routing and generation stages so intent classification decided which pipeline a request entered before any content was produced.
- **Advanced RAG & Latency Optimisation:** Independently architected a latency- and cost-optimised, session-based RAG pipeline with Redis-backed session memory over ChromaDB, serving as a knowledge retrieval system across Voice Bot and Agent Assist products, with an ingestion pipeline processing large volumes of unstructured enterprise documents into the retrieval layer. Held retrieval inside a two-second budget for live conversation, mitigated hallucination, and tuned what context reached the model at each turn rather than passing raw top-k results, with a feedback loop that fed real usage back into the pipeline to refine retrieval quality and reduce failure rates.
- **Real-Time Voice AI & Conversational Logic:** Engineered end-to-end voice bot pipelines (STT to LLM to TTS) optimised for real-time interaction and concurrent call handling, with intent detection running as its own microservice and a state-machine routing engine handling topic switching, context management and fallback, using streaming to hold responses under two seconds, with escalation to human agents when the bot could not resolve a request.
- **Data Privacy & LLM Security:** Built an automated ML pipeline using Microsoft Presidio and Hugging Face NER models to mask Personally Identifiable Information in live conversation text before it reached the LLM and de-mask it before the response returned to TTS, keeping the system GDPR compliant without breaking the conversation.
- **Workflow Automation (Textflow):** Built and shipped an end-to-end email automation product on the Azure Outlook Client API, putting autonomous analysis, contextual drafting and dispatch through LLM processing into daily use by business teams.

Generative AI Intern and Work Student, Sogedes GmbH

04/2023 – 11/2024 | Mannheim, Germany

- **Semantic Search:** Implemented foundational semantic search capabilities for efficient query retrieval from vector databases, setting the technical stage for future RAG architectures.
- **System Modernisation:** Modernised enterprise chatbot and voice bot systems by integrating modern LLMs into legacy rule-based frameworks (RASA, DialogFlow CX), moving to hybrid architectures that keep deterministic flows where they are reliable and use LLM reasoning where they are not, improving intent recognition, fallback handling and response quality. Conducted platform evaluations including Cognigy.

Data Analytics Intern, Netlink Software Group America Inc

02/2021 – 05/2021 | Bhopal, India

- NLP tasks with Python libraries such as Spacy and NLTK.

🎓 EDUCATION

M.Sc in Artificial Intelligence and Data Science

03/2022 – 09/2025 | Deggendorf, Germany

Technische Hochschule Deggendorf

- **Joint Degree Program with the University of South Bohemia, Czech Republic**
- **Majoring Courses:** Advanced Machine Learning, Theoretical Fundamentals of AI, Advanced Topics in AI, Distributed Algorithms, Advanced Data Storages and Analyses
- **Grade:** 2.2(Good) - German scale 1.0 (highest) to 5.0

SKILLS

Core AI & Agentic — LangGraph, LangChain, DeepAgents, Tool Calling and Function Calling, Agent Harness Design, Multi-Agent Systems, LLM Orchestration, Advanced RAG, Vector Search, Prompt and Context Engineering, Model Context Protocol (MCP), State and Memory Management, Guardrails, Context Engineering and Compression

Cloud & DevOps — Docker and container-based deployments, Microsoft Azure AI (AI Services, AI Search), Google Cloud (Vertex AI, GCS), CI/CD (GitHub Actions), Git/GitHub

Backend & Data — Python (Async), FastAPI, Pydantic, REST APIs, Microservices and Distributed Services, SQL (PostgreSQL), Redis, Unicorn, Vector Databases (ChromaDB), Data Pipeline Design and Integration, Unstructured Document Processing

Machine Learning & NLP — PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, NLP, NLU, NER, Intent Classification, PII Extraction and Masking, Generative Modelling (GAN, TimeGAN)

Evaluation & Reliability — AI Evaluation (accuracy, latency, cost), Evaluation Methodology, Feedback Loops, Monitoring and Observability, Responsible AI and Data Governance

Models & Platforms — OpenAI (ChatGPT), Anthropic (Claude), Gemini, Groq, HuggingFace, Claude Code, Vercel AI SDK, RASA, DialogFlow CX, Cognigy (Evaluated)

Frontend — JavaScript, Java, TypeScript, Node.js, React

Agile/Scrum — Jira, Kanban

Languages — English (Fluent), German (A2)

PROJECTS

Local Developer OS (MCP Server Integration)

- Architected a custom Model Context Protocol (MCP) server in Python to securely expose local file systems and Git environments to LLM clients.
- Engineered secure tool-calling capabilities, enabling context-aware AI agents to autonomously read system logs, navigate repositories, and assist in local debugging.

Automated PR Reviewer & Security Scanner

- Developed a DevOps automation service using FastAPI, PyGithub, and GitHub Actions that intercepts webhooks to analyse pull requests automatically.
- Integrated the Gemini API to review code diffs, identify security vulnerabilities, and autonomously post contextual feedback, streamlining the CI/CD workflow.

Empowering Smart Meter Applications: GAN-Generated Synthetic Data for Water and Power Analysis

- Developed a GAN-based model to generate synthetic data for water and power consumption analysis.
- Handled the full training workflow, including data preparation, feature engineering, model training and evaluation, using Python, TensorFlow and TimeGAN, ensuring data privacy and usability for smart meter applications.

INTERESTS

Gaming & Game Dev — Passionate about exploring games with cool mechanics and interactive storytelling, building on my background in Unity3D/C#.

Travel, Music, and Sports